# CURRICULUM VITAE — ROLAND KWITT



| | | |
|---|---|---|
| *Address* | **Abtsdorf 145, Attersee am Attersee** | ✉ Roland.Kwitt@plus.ac.at |
| *Birth Date* | March 19, 1982 | |
| *Citizenship* | Austrian | |
| *Academic Details* | *h*-index: 39, Cites: 6134 (Source: Google Scholar, 04/2026) | |
| *Webpage* | https://rkwitt.github.io/ | |
| *DBLP* | https://dblp.org/pers/hd/k/Kwitt:Roland | |
| *ORCID* | https://orcid.org/0000-0001-9947-4465 | |



## CURRENT EMPLOYMENT

2022 - now **University of Salzburg**
*Full Professor (for Machine Learning, § 99 (4) UG)*
Department of Artificial Intelligence and Human Interfaces (AIHI)
Jakob-Haringer Str. 2, A-5020 Salzburg, Austria

2020 - 2021 **University of Salzburg**
*Full Professor (for Machine Learning, § 99 (4) UG)*
Department of Computer Science
Jakob-Haringer Str. 2, A-5020 Salzburg, Austria

2017 - 2020 **University of Salzburg**
*Associate Professor*
Department of Computer Science
Jakob-Haringer Str. 2, A-5020 Salzburg, Austria

2013 - 2017 **University of Salzburg**
*Assistant Professor*
Department of Computer Science
Jakob-Haringer Str. 2, A-5020 Salzburg, Austria

## PREVIOUS EMPLOYERS

2011 - 2013 **Kitware Inc.**
*R & D Engineer*, Computer Vision / Medical Imaging Group
101 E Weaver St., NC 27510, USA
Supervisor(s): Stephen Aylward, Brad Davis

## EDUCATION

2010 - 2011 *PostDoc*, CS department, University of Salzburg (ADVISOR(S): Andreas Uhl, Wolfgang Pree)
2007 - 2010 *Dr. techn. (equiv. to PhD)*, CS department, University of Salzburg (ADVISOR: Andreas Uhl)
2005 - 2007 *Dipl.-Ing. (equiv. to MsC)*, CS department, University of Salzburg (ADVISOR: Ulrich Hofman)
2001 - 2005 *Dipl.-Ing. (FH) (equiv. MsC)*, Telecommunications Engineering, University of Applied Sciences Salzburg (ADVISOR: Ulrich Hofman)

## AWARDS

2014 **CVPR '14 Outstanding Reviewer**
2012 Short-listed for the "Heinz-Zemanek" price 2012 (notification e-mail upon request)
2012 **MICCAI '12 Young Investigator Award**, awarded at MICCAI '12 (Nice, France)
2007 **Best Paper Award**, International Conference on Computer Recognition Systems (CORES '07)
2005 **Special Appreciation Award**, Austrian Ministry of Science and Research

## Main Research Areas

**Machine learning**, Computer vision, Medical image analysis

## Ranking in selection processes for *Full Professorships*

- Ranked **1st** for § 98 UG professorship *Data Science* at the University of Innsbruck ("Ruf" received 12/2019)
- Ranked **1st** for § 99 UG professorship *Machine Learning / Data Science* at the University of Klagenfurt ("Ruf" received 06/2020)

## Third-Party Funding (granted)

2025-2028 *doc.funds.connect REVELATION: Artificial Intelligence driven Biomedical Imaging Innovation* (granted)
(joint project with the FH Salzburg as project lead; 6 PhD positions)
Funding source: FWF & Land Salzburg
Project volume € 1.152.216,50.-

2023-2025 *servEB: Globales Daten- und Patientenmanagement bei seltenen, genetischen Hautkrankheiten* (ongoing)
(joint project with the Paracelsus Medical University and the Salzburger Landeskliniken (Lead))
Funding source: Amt der Salzburger Landesregierung
Project volume € 478.000,00.- (University of Salzburg share: € 100.000,00.-)

2023 - 2027 *Intelligent Data Analytics (IDA) Lab III* (ongoing)
(together with W. Trutschnig (Head), C. Borgelt, and A. Bathke, Department of Artificial Intelligence and Human Interfaces, University of Salzburg)
Funding source: Amt der Salzburger Landesregierung (within the WISS[1] initiative)
Project volume € 1,600.000,00.-

2019 - 2023 *Intelligent Data Analytics (IDA) Lab I & II*
(together with W. Trutschnig (Head), Department of Mathematics, University of Salzburg and C. Borgelt, Department of Mathematics/Computer Science, University of Salzburg)
Funding source: Amt der Salzburger Landesregierung (within the WISS[1] initiative)
Project volume € 2,300.000,00.-

2020 - 2022 *Free-of-Bias, Robust and Intelligent Data Analytics (FRIDA)* (completed)
(together with W. Trutschnig, Department of Mathematics, University of Salzburg)
Funding source: Porsche Holding GmBH
Industry partner: Porsche Informatik GmbH
Project volume: € 250.000,00.-

2019 - now *Deep Homological Learning* (completed)
Funding source: FWF (Project Nr. `P 31799`)
Project volume: € 238.512,75

2019 - 2022 *Kundenfokussierte Zukunftstrends (KFZ)* (completed)
(together with W. Trutschnig, Department of Mathematics, University of Salzburg)
Funding source: Land Salzburg (within the WISS 2025 initiative)
Industry partner: Porsche Informatik GmbH
Project volume: € 478.610,38.-

2018 - 2021 *"Kleinprojekte" Critical data & Feature selection* (completed)
(together with W. Trutschnig, Department of Mathematics, University of Salzburg)
Funding source: Porsche Informatik GmbH
Industry partner: Porsche Informatik GmbH
Project volume: € 40,000.-

2018 *Synonym Analysis for Improving Search Queries* (completed)
(together with N. Augsten, Department of Computer Science, University of Salzburg)

[1] Wissenschafts- und Innovationsstrategie Salzburg

Industry partner: FindoLogic GmbH
Funding source: FFG (Innovationsscheck 5,000)
Project volume: € 5,000.-

2018 *Data Analytics in Industrial Environments* (completed)
(together with W. Trutschnig, Department of Mathematics, University of Salzburg)
Industry partner: Siemens Austria
Project volume: € 20,000.-

## Student Supervision

### Primary PhD/PostDoc advisor

- Dominik Geng (PhD student, since 2025)
- Martin Uray (PhD student, since 2024, co-supervised with Stefan Huber)
- Günther Eder (PostDoc completed, now in industry)
- Sebastian Zeng (PhD completed, now PostDoc at Linnaeus University, Sweden)
- Florian Graf (PhD completed, now PostDoc in my group)
- Christoph D. Hofer (PhD & PostDoc completed, now in industry)

### Secondary PhD advisor

- Mann Willi (PhD completed, now in industry)
- Wimmer Georg (PhD completed, now PostDoc at University of Salzburg)
- Kauba Christof (PhD completed, now PostDoc at University of Salzburg)
- Debiasi Luca (PhD completed, now in industry)
- Ribeiro Eduardo (PhD completed)
- Schraml Rudolf (PhD completed)
- Kirchgasser Simon Ignaz (PhD ongoing)
- González Tejeda Yansel (PhD ongoing)

### MSc advisor

- Johanna Wald (graduated 2017)
- Schmitzberger Nina Marie (graduated 2020)
- Michael Kastner (graduated 2020)
- Söllinger Dominik (graduated 2020)
- Tobias Hilgart (graduated 2020)
- Philip Brandauer (graduated 2020)
- Grafendorfer Philipp (graduated 2021)
- Peer Raphael (graduated 2021)
- Marlene Holzleitner (graduated 2021)
- Sabine Hasenleithner (graduated 2026)

## Publications (in reverse-chronological order)

For conference articles, the (most recent, i.e., 2026) ICORE ranking is provided as **A*, A, B,...** (if available). Similarly, for journal articles, the journal impact factor (**JIF**) is listed as well (from the Clarivate Master Journal List from 2024). Also, if available, the *Digital Object Identifier (DOI)* is provided via a clickable doi symbol.

### Journal Article (peer-reviewed)


[Dyc+21] L.E. van Dyck, R. Kwitt, S.J. Denzler, and W.R. Gruber. "Comparing object recognition in humans and deep convolutional neural networks - An eye tracking study". In: *Front. Neurosci* 15 (2021). doi. **JIF: 3.2**

[C H+20] C. Hofer, R. Kwitt, Y. Höller, E. Trinka, and A. Uhl. "An empirical assessment of appearance descriptors applied to MRI for automated diagnosis of TLE and MCI". In: *Comput. Biol. Med.* 117 (2020). doi. **JIF: 6.3**

[CRM19] C. Hofer, R. Kwitt, and M. Niethammer. "Learning Representations of Persistence Barcodes". In: *J. Mach. Learn. Res.* 20.126 (2019), pp. 1–45. **JIF: 5.2**

[D R+19] D. R. Chittajallu, M. McCormick, S. Gerber, T.J. Czernuszewicz, R. Gessner, M.S. Willis, M. Niethammer, R. Kwitt, and S.R. Aylward. "Image-Based Methods for Phase Estimation, Gating, and Temporal Superresolution of Cardiac Ultrasound". In: *IEEE Trans. Biomed. Eng.* 66.1 (2019), pp. 72–79. doi. **JIF: 4.5**

[N S+19] N. Stanley, T. Bonacci, R. Kwitt, M. Niethammer, and P.J. Mucha. "Stochastic Block Models with multiple continuous attributes". In: *Appl. Netw. Sci.* 4.54 (2019). doi. **JIF: 1.5**

[Z D+19] Z. Ding, G. Fleishman, X. Yang, P. Thomson, R. Kwitt, and M. Niethammer. "Fast predictive simple geodesic regression". In: *Med. Image Anal.* 56 (2019), pp. 193–209. doi. **JIF: 11.8**

[N S+18] N. Stanley, R. Kwitt, M. Niethammer, and P.J. Mucha. "Compressing Networks with Super Nodes". In: *Nature Sci. Rep.* 8.10892 (2018). doi.

[X H+18] X. Han, R. Kwitt, S.R. Aylward, S. Bakas, B. Menze, A. Asturias, P. Vespa, J. van Horn, and M. Niethammer. "Brain extraction from normal and pathological images: A joint PCA/Image-Reconstruction approach". In: *NeuroImage* 176.8 (2018), pp. 431–445. doi. **JIF: 4.5**

[Yan+17] X. Yang, R. Kwitt, M. Styner, and M. Niethammer. "Quicksilver: Fast Predictive Image Registration - a Deep Learning Approach". In: *NeuroImage* 158 (2017), pp. 378–396. doi. **JIF: 4.5**

[Hon+16a] Y. Hong, R. Kwitt, N. Singh, N. Vasconcelos, and M. Niethammer. "Parametric Regression on the Grassmannian". In: *IEEE Trans. Pattern Anal. Mach. Intell.* 38.11 (2016). doi. **JIF: 18.6**

[Liu+15a] X. Liu, M. Niehthammer, R. Kwitt, N. Singh, M. McCormick, and S. Aylward. "Low-Rank Atlas Image Analyses in the Presence of Pathologies". In: *IEEE Trans. Med. Imaging* 34.12 (2015), pp. 2583–2591. doi. **JIF: 9.8**

[Hon+14c] Y. Hong, B. Davis, J. S. Marron, R. Kwitt, N. Singh, J. S. Kimbell, E. Pitkina, R. Superfine, S.D. Davis, C. J. Zdanski, and M. Niethammer. "Statistical atlas construction via weighted functional boxplots". In: *Med. Image Anal.* 18.4 (2014), pp. 684–698. doi. **JIF: 11.8**

[Kwi+13b] R. Kwitt, N. Vasconcelos, S. Razzaque, and S. Aylward. "Localizing Target Structures in Ultrasound Video - A Phantom Study". In: *Med. Image Anal.* 17.7 (2013), pp. 712–722. doi. **JIF: 11.8**

[Kwi+12b] R. Kwitt, N. Vasconcelos, N. Rasiwasia, A. Uhl, B. Davis, M. Häfner, and F. Wrba. "Endoscopic Image Analysis in Semantic Space". In: *Med. Image Anal.* 16.7 (2012), pp. 1415–1422. doi. **JIF: 11.8**

[KMU11a] R. Kwitt, P. Meerwald, and A. Uhl. "Efficient Texture Image Retrieval Using Copulas in a Bayesian Framework". In: *IEEE Trans. Image Process.* 20.7 (2011), pp. 2063–2077. doi. **JIF: 13.7**

[KMU11b] R. Kwitt, P. Meerwald, and A. Uhl. "Lightweight Detection of Additive Watermarking in the DWT-Domain". In: *IEEE Trans. Image Process.* 20.2 (2011), pp. 474–484. doi. **JIF: 13.7**

[KU10a] R. Kwitt and A. Uhl. "Lightweight Probabilistic Texture Retrieval". In: *IEEE Trans. Image Process.* 19.1 (2010), pp. 241–253. doi. **JIF: 13.7**

[Haf+09a] M. Häfner, R. Kwitt, A. Uhl, A. Gangl, F. Wrba, and A. Vécsei. "Feature-Extraction from Multi-Directional Multi-Resolution Image Transformations for the Classification of Zoom-Endoscopy Images". In: *Pattern Anal. Appl.* 12.4 (2009), pp. 407–413. . **JIF: 2.0**

[Haf+08a] M. Häfner, R. Kwitt, A. Uhl, A. Gangl, F. Wrba, and A. Vécsei. "Computer-assisted Pit-Pattern Classification in Different Wavelet Domains for Supporting Dignity Assessment of Colonic Polyps". In: *Pattern Recognit.* 42.6 (2008), pp. 1180–1191. . **JIF: 7.6**

## Conference Articles (peer-reviewed)

[F G+25] F. Graf, P. Pellizzoni, M. Uray, S. Huber, and R. Kwitt. "The Flood Complex: Large-Scale Persistent Homology on Millions of Points". In: *NeurIPS*. 2025. **A***

[Gre+25] H. Greer, L. Tian, F.-X. Vialard, R. Kwitt, R.S.J. Estepar, and M. Niethammer. "CARL: A Framework for Equivariant Image Registration". In: *CVPR*. 2025. **A***

[Dem+24] B. Demir, L. Tian, H. Greer, R. Kwitt, F.-X. Vialard, R.S.J. Estepar, S. Bouix, R. Rushmore, E. Ebrahim, and M. Niethammer. "MultiGradICON: A Foundation Model for Multimodal Medical Image Registration". In: *WBIR*. 2024. .

[L T+24] L. Tian, H. Greer, R. Kwitt, F.-X. Vialard, R.S.J. Estepar, S. Bouix, R. Rushmore, and M. Niethammer. "uniGradICON: A Foundation Model for Medical Image Registration". In: *MICCAI*. 2024. **A**

[Pap+24] T. Papamarkou, T. Birdal, M. Bronstein, G. Carlsson, J. Curry, Y. Gao, M. Hajij, R. Kwitt, P. Lio, P. Di Lorenzo, V. Maroulas, N. Miolane, F. Nasrin, K.N. Ramamurthy, B. Rieck, S. Scardapane, M.T. Schaub, P. Velickovic, B. Wang, Y. Wang, G.-W. Wei, and G. Zamzmi. "Position Paper: Challenges and Opportunities in Topological Deep Learning". In: *ICML*. 2024. **A***

[Zen+24] S. Zeng, F. Graf, M. Uray, S. Huber, and R. Kwitt. "Neural Persistence Dynamics". In: *NeurIPS*. 2024. **A***

[Gre+23] H. Greer, L. Tian, F.-X. Vialard, R. Kwitt, S. Bouix, Estepar R.S.J., R.J. Rushmore, and Marc Niethammer. "Inverse Consistency by Construction for Multistep Deep Registration". In: *MICCAI*. 2023. . **A**

[Tia+23] L. Tian, H. Greer, F.-X. Vialard, Estepar R.S.J., R.J. Rushmore, N. Makris, S. Bouix, and Marc Niethammer. "GradICON: Approximate Diffeomorphisms via Gradient Inverse Consistency". In: *CVPR*. 2023. **A***

[ZGK23] Sebastian Zeng, Florian Graf, and Roland Kwitt. "Latent SDEs on Homogeneous Spaces". In: *NeurIPS*. 2023. **A***

[Gra+22] F. Graf, S. Zeng, B. Rieck, M. Niethammer, and R. Kwitt. "On Measuring Excess Capacity in Neural Networks". In: *NeurIPS*. 2022. **A***

[F G+21] F. Graf, C. Hofer, M. Niethammer, and R. Kwitt. "Dissecting Supervised Constrastive Learning". In: *ICML*. 2021. **A***

[H G+21] H. Greer, R. Kwitt, F.-X. Vialard, and M. Niethammer. "ICON: Learning Regular Maps Through Inverse Consistency". In: *ICCV*. 2021. . **A***

[ZGK21] S. Zeng, F. Graf, and C. Hofer R. Kwitt. "Topological Attention for Time Series Forecasting". In: *NeurIPS*. 2021. **A***

[C H+20a] C. Hofer, F. Graf, B. Rieck, M. Niethammer, and R. Kwitt. "Graph Filtration Learning". In: *ICML*. 2020. **A***

[C H+20b] C. Hofer, F. Graf, M. Niethammer, and R. Kwitt. "Topologically Densified Distributions". In: *ICML*. 2020. **A***

[F-X+20] F.-X. Vialard, R. Kwitt, S. Wei, and M. Niethammer. "A Shooting Formulation of Deep Learning". In: *NeurIPS*. 2020. **A***

[C H+19] C. Hofer, R. Kwitt, M. Dixit, and M. Niethammer. "Connectivity-Optimized Representation Learning via Persistent Homology". In: *ICML*. 2019. **A***

[MF19] M. Niethammer and R. Kwitt F.-X. Vialard. "Metric Learning for Image Registration". In: *CVPR*. 2019. . **A***

[Liu+18a] B. Liu, M. Dixit, R. Kwitt, and N. Vasconcelos. "Feature Space Transfer for Data Augmentation". In: *CVPR*. 2018. . A*

[Gre+18] H. Greer, S. Gerber, M. Niethammer, R. Kwitt, M. McCormick, D. Chittajallu, N. Siekierski, M. Oetgen, K. Cleary, and S. Aylward. "Scoliosis Screening and Monitoring Using Self Contained Ultrasound and Neural Networks". In: *ISBI*. 2018. .

[Dix+17] M. Dixit, R. Kwitt, M. Niethammer, and N. Vasconcelos. "AGA: Attribute-Guided Augmentation". In: *CVPR*. 2017. . A*

[Han+17] X. Han, X. Yang, R. Kwitt, and M. Niethammer. "Efficient Registration of Pathological Images: A joint PCA/Image-Reconstruction Approach". In: *ISBI*. 2017. .

[Hof+17a] C. Hofer, R. Kwitt, Y. Höller, E. Trinka, M. Niethammer, and A. Uhl. "Constructing Shape Spaces from a Topological Perspective". In: *IPMI*. 2017. .

[Hof+17b] C. Hofer, R. Kwitt, Y. Höller, E. Trinka, and A. Uhl. "Simple Domain Adaptation for Cross-Dataset Analyses of Brain MRI Data". In: *ISBI*. 2017. .

[Hof+17c] C. Hofer, R. Kwitt, M. Niethammer, and A. Uhl. "Deep Learning with Topological Signatures". In: *NIPS*. 2017. A*

[Hon+17] Y. Hong, X. Yang, R. Kwitt, M. Styner, and M. Niethammer. "Regression Uncertainty on the Grassmannian". In: *AISTATS*. 2017. A

[Yan+17] X. Yang, R. Kwitt, M. Styner, and M. Niethammer. "Fast Predictive Multimodal Image Registration". In: *ISBI*. 2017. .

[Gad+16a] M. Gadermayr, S. Hegenbart, R. Kwitt, and A. Uhl. "Narrow Band Imaging Versus White-Light: What is best for Computer-Assisted Diagnosis of Celiac Disease?" In: *ISBI*. 2016. .

[KHN16a] R. Kwitt, S. Hegenbart, and M. Niethammer. "One-Shot Learning of Scene Locations via Feature Trajectory Transfer". In: *CVPR*. 2016. .

[Ayl+16a] S. Alyward, M. McCormick, H.J. Kang, S. Razzaque, R. Kwitt, and M. Niethammer. "Ultrasound Spectroscopy". In: *ISBI*. 2016. .

[Yan+16] X. Yang, X. Han, E. Park, S. Aylward, R. Kwitt, and M. Niethammer. "Registration of Pathological Images". In: *Proceedings of the MICCAI Workshop on Simulation and Synthesis in Medical Imaging*. 2016. .

[YKN16] X. Yang, R. Kwitt, and M. Niethammer. "Fast Predictive Image Registration". In: *Proceedings of the MICCAI Workshop on Deep Learning in Medical Image Analysis*. 2016. .

[Kwi+15a] R. Kwitt, S. Huber, M. Niethammer, W. Lin, and U. Bauer. "Statistical Topological Data Analysis – A Kernel Perspective". In: *NIPS*. 2015.

[Rei+15a] R. Reininghaus, U. Bauer, S. Huber, and R. Kwitt. "A Stable Multi-scale Kernel for Topological Machine Learning". In: *CVPR*. 2015. . A*

[Hon+15a] Y. Hong, N. Singh, R. Kwitt, and M. Niethammer. "Group Testing for Longitudinal Data". In: *IPMI*. 2015. .

[HKN15a] Y. Hong, R. Kwitt, and M. Niethammer. "Model Criticism for Regression on the Grassmannian". In: *MICCAI*. 2015. . A

[Hon+14a] Y. Hong, N. Singh, R. Kwitt, and M. Niethammer. "Time-warped Geodesic Regression". In: *MICCAI*. 2014. . A

[Kwi+14a] R. Kwitt, S. Razzaque, J. Lowell, and S. Aylward. "Variability sensitivity of dynamic texture based recognition in clinical CT data". In: *SPIE Medical Imaging*. 2014. .

[Liu+14a] X. Liu, M. Niethammer, R. Kwitt, M. McCormick, and S. Aylward. "Low-Rank to the Rescue: Atlas-based Analyses in the Presence of Pathologies". In: *MICCAI*. 2014. . A

[Heg+14a] S. Hegenbart, R. Kwitt, N. Rasiwasia, A. Vécsei, and A. Uhl. "Do We need Annotation Experts? A Case Study in Celiac Disease Classification". In: *MICCAI*. 2014. . A

[Hon+14b] Y. Hong, R. Kwitt, N. Singh, B. Davis, and M. Niethammer. "Geodesic Regression on the Grassmannian". In: *ECCV*. 2014. .

[Hon+13a] Y. Hong, B. Davis, J.S. Marron, R. Kwitt, and M. Niethammer. "Weighted Functional Boxplot with Application to Statistical Atlas Construction". In: *MICCAI*. 2013. . A

[Kwi+13a] R. Kwitt, D. Pace, M. Niethammer, and S. Aylward. "Studying Cerebral Vasculature Using Structure Proximity and Graph Kernels". In: *MICCAI*. 2013. . **A**

[KVR12a] R. Kwitt, N. Vasconcelos, and N. Rasiwasia. "Scene Recognition on the Semantic Manifold". In: *ECCV*. 2012. .

[Kwi+12a] R. Kwitt, N. Vasconcelos, S. Razzaque, and S. Alyward. "Recognition in Ultrasound Videos: Where Am I?" In: *MICCAI*. 2012. . **A**

[Gsc+11a] M. Gschwandtner, R. Kwitt, W. Pree, and A. Uhl. "Infrared Camera Calibration for Dense Depth Map Construction". In: *IV*. 2011. .

[GKU11a] M. Gschwandtner, R. Kwitt, and A. Uhl. "BlenSor: Blender Sensor Simulation Toolbox". In: *ISVC*. 2011. .

[Kwi+11b] R. Kwitt, P. Meerwald, A. Uhl, and G. Verdoolaege. "Testing a Multivariate Model for Wavelet Coefficients". In: *ICIP*. 2011. . **A**

[Kwi+11a] R. Kwitt, N. Rasiwasia, N. Vasconcelos, A. Uhl, M. Häfner, and F. Wrba. "Learning Pit Pattern Concepts for Gastroenterological Training". In: *MICCAI*. 2011. . **A**

[Hub+10a] S. Huber, R. Kwitt, P. Meerwald, M. Held, and A. Uhl. "Watermarking of 2D Vector Graphics with Distortion Constraint". In: *ICME*. 2010. .

[Kwi+10a] R. Kwitt, A. Uhl, M. Häfner, A. Gangl, F. Wrba, and A. Vécsei. "Predicting the Histology of Colorectal Lesions in a Probabilistic Framework". In: *MMBIA*. 2010. .

[Haf+09b] M. Häfner, A. Gangl, R. Kwitt, A. Uhl, A. Vécsei, and F. Wrba. "Improving Pit-Pattern Classification of Endoscopy Images by a Combination of Experts". In: *MICCAI*. 2009. . **A**

[Heg+09c] S. Hegenbart, R. Kwitt, M. Liedlgruber, A. Uhl, and A. Vécsei. "Impact of Duodenal Image Capturing Techniques and Duodenal Regions on the Performance of Automated Diagnosis of Celiac Disease". In: *ISPA*. 2009. .

[KMU09d] R. Kwitt, P. Meerwald, and A. Uhl. "A Joint Model of Complex Wavelet Coefficients for Texture Retrieval". In: *ICIP*. 2009. . **B**

[KMU09c] R. Kwitt, P. Meerwald, and A. Uhl. "Efficient Detection of Additive Watermarking in the DWT-Domain". In: *EUSIPCO*. 2009.

[KMU09b] R. Kwitt, P. Meerwald, and A. Uhl. "Blind DT-CWT Domain Additive Spread-Spectrum Watermark Detection". In: *DSP*. 2009. .

[KMU09a] R. Kwitt, P. Meerwald, and A. Uhl. "Color-Image Watermarking using Multivariate Power-Exponential Distribution". In: *ICIP*. 2009. . **B**

[Haf+08b] M. Häfner, R. Kwitt, F. Wrba, A. Gangl, A. Vécsei, and A. Uhl. "One-Against-One Classification for Zoom-Endoscopy Images". In: *MEDSIP*. 2008. .

[KU08b] R. Kwitt and A. Uhl. "Color Eigen-Subband Features for Endoscopy Image Classification". In: *ICASSP*. 2008. .

[KU08a] R. Kwitt and A. Uhl. "Image Similarity Measurement by Kullback-Leibler Divergences between Complex Wavelet Subband Statistics for Texture Retrieval". In: *ICIP*. 2008. . **B**

[KU07a] R. Kwitt and A. Uhl. "Modeling the Marginal Distributions of Complex Wavelet Coefficient Magnitudes for the Classification of Zoom-Endoscopy Images". In: *MMBIA*. 2007. .

## Theses

[Kwitt10a] R. Kwitt. "Statistical Modeling in the Wavelet Domain and Applications". PhD thesis. Department of Computer Science, University of Salzburg, Austria, 2010.

## Conference Talks & Presentations

12/2025 *The Flood Complex: Large-Scale Persistent Homology on Millions of Points*, NeurIPS '25, San Diego, USA
12/2024 *Neural Persistence Dynamics*, NeurIPS '24, Vancouver, CA
12/2023 *Latent SDEs on Homogeneous Spaces*, NeurIPS '23, New Orleans, USA
12/2022 *On Measuring Excess Capacity in Neural Networks*, NeurIPS '22, New Orleans, USA
12/2021 *Topological Attention for Time Series Forecasting*, NeurIPS '21 (Virtual conference)
06/2021 *Dissecting Supervised Contrastive Learning*, ICML '21 (Virtual conference)
12/2020 *A Shooting Formulation of Deep Learning*, NeurIPS '20 (Virtual conference)
07/2020 *Topologically Densified Distributions*, ICML '20, Vienna, Austria (Virtual conference)
07/2020 *Graph Filtration Learning*, ICML '20, Vienna, Austria (Virtual conference)
09/2019 *Deep Homological Learning*, ÖMG Tagung, Dornbirn, Austria
06/2019 *Metric Learning for Image Registration*, CVPR '19, Long Beach, CA, USA
06/2016 *One-Shot Learning of Scene Locations via Feature Trajectory Transfer*, CVPR '16, Las Vegas, USA
12/2015 *Statistical Topological Data Analysis – A Kernel Perspective*, NIPS '15, Montreal, Canada
10/2015 *Model Criticism for Regression on the Grassmannian*, MICCAI '15, Munich, Germany
06/2015 *A Stable Multi-Scale Kernel for Topological Machine Learning*, CVPR '15, Boston, USA
09/2014 *Geodesic Regression on the Grassmannian*, ECCV '14, Zurich, Switzerland
09/2014 *Do we need Annotation Experts – A Case Study in Celiac Disease Classification*, MICCAI '14, Boston, USA
09/2014 *Low-Rank to the Rescue – Atlas-based Analyses in the Presence of Pathologies*, MICCAI '14, Boston, USA
10/2013 *Studying Cerebral Vasculature Using Structure Proximity and Graph Kernels*, MICCAI '13, Nagoya, Japan
10/2012 *Scene Recognition on the Semantic Manifold*, ECCV '12, Florence, Italy
10/2012 *Recognition in US Video: Where Am I?*, MICCAI '12, Nice, France
09/2011 *Learning Pit Pattern Concepts for Gastroenterological Training*, MICCAI '11, Toronto, Canada
08/2010 *Statistical Modeling in the Wavelet Domain and Applications*, PhD defense, Salzburg, Austria
11/2009 *A Joint Model of Complex Wavelet Coefficients for Texture Retrieval*, ICIP '09, Cairo, Egypt
11/2009 *Color-Image Watermarking using Multivariate Power-Exponential Distribution*, ICIP '09, Cairo, Egypt
09/2009 *Improving Pit Pattern Classification by a Combination of Experts*, MICCAI '09, London, UK
10/2008 *Image Similarity Measurement by Kullback-Leibler Divergences between Complex Wavelet Subband Statistics for Texture Retrieval*, ICIP '08, San Diego, CA, USA
04/2008 *Color Eigen-Subband Features for Endoscopy Image Classification*, ICASSP '08, Las Vegas, NV, USA
10/2007 *Modeling the Marginal Distributions of Complex Wavelet Coefficient Magnitudes for the Classification of Zoom-Endoscopy Images*, MMBIA '07, Rio de Janeiro, Brazil

## Invited Talks

04/2022 *Topologically Densified Distributions*
ICLR Workshop on Geometrical and Topological Representation Learning 2022 (Virtual)
09/2021 *Topological Machine Learning*
Thematic Mini-Conference on Computational Topology and Machine Learning (Virtual)
Berlin Mathematics Research Center
09/2019 *Deep Homological Learning*
ÖMG Conference 2019, Dornbirn (in the *Computational Geometry and Topology* session)
01/2018 *Machine Learning with Topological Signatures*
Oberwolfach Workshop "Statistics for Data with Geometric Structure", Oberwolfach, Germany
04/2016 *Low rank to the Rescue: Atlas-based Analyses in the Presence of Pathologies*
"Images and Networks of the Brain", Hamburg, Germany
07/2015 *Topological Machine Learning*
ISNPS '15, Graz Austria
04/2014 *Grassmannian Geodesic Regression*
IST Austria, Austria
06/2013 *Localizing Target Structures In Ultrasound Videos*
Quantitative Medical Imaging (QMI), Arlington, VA, USA
12/2012 *Scene Recognition on the Semantic Manifold*
SVCL, UC San Diego, USA
10/2012 *Recognition in US Video: Where Am I?*
University of North Carolina, Chapel Hill (Computer Science), NC, USA

## Teaching

*Except for a small selection of undergraduate courses, all lectures/labs are taught in English.*

| | |
|---|---|
| SS 26 | Seminar Multimedia Technologies (SE, graduate level), University of Salzburg |
| SS 26 | Statistical Learning Theory (VO+PS, graduate level), University of Salzburg |
| WS 25/26 | AI Lab / AI Werkstatt (UV+SE, undergraduate level), University of Salzburg |
| WS 25/26 | Machine Learning (VO, undergraduate level), University of Salzburg |
| WS 25/26 | Introduction to AI / AI Eingangswerkstatt (VO, undergraduate level), University of Salzburg |
| WS 25/26 | Computer Vision (VO+PS, graduate level), University of Salzburg |
| SS 25 | Statistical Learning Theory (VO+PS, graduate level), University of Salzburg |
| SS 25 | Seminar Multimedia Technologies (SE, graduate level), University of Salzburg |
| WS 24/25 | AI Lab / AI Werkstatt (UV+SE, undergraduate level), University of Salzburg |
| WS 24/25 | Machine Learning (VO, undergraduate level), University of Salzburg |
| WS 24/25 | Introduction to AI / AI Eingangswerkstatt (VO, undergraduate level), University of Salzburg |
| WS 24/25 | Interpreting and Presenting Statistical Analyses (SE, graduate level), University of Salzburg |
| WS 24/25 | Case Studies (SE, graduate level), University of Salzburg |
| WS 24/25 | Computer Vision (VO+PS, graduate level), University of Salzburg |
| SS 24 | Statistical Learning Theory (VO+PS, graduate level), University of Salzburg |
| SS 24 | Seminar Multimedia Technologies (SE, graduate level), University of Salzburg |
| WS 23/24 | Machine Learning (VO, undergraduate level), University of Salzburg |
| WS 23/24 | Introduction to AI / AI Eingangswerkstatt (VO, undergraduate level), University of Salzburg |
| WS 23/24 | Interpreting and Presenting Statistical Analyses (SE, graduate level), University of Salzburg |
| WS 23/24 | Case Studies (SE, graduate level), University of Salzburg |
| WS 23/24 | Seminar Multimedia Technologies (SE, graduate level), University of Salzburg |
| WS 23/24 | Computer Vision (VO+PS, graduate level), University of Salzburg |
| SS 23 | Statistical Learning Theory (VO+PS, graduate level), University of Salzburg |
| SS 23 | Databases 1 (VO+PS, undergraduate level), University of Salzburg |
| SS 23 | Medical Imaging (VO+PS, graduate level), University of Salzburg |
| SS 23 | Seminar Multimedia Technologies (SE, graduate level), University of Salzburg |
| WS 22/23 | Introduction to AI / AI Eingangswerkstatt (VO, undergraduate level), University of Salzburg |
| WS 22/23 | Interpreting and Presenting Statistical Analyses (SE, graduate level), University of Salzburg |
| WS 22/23 | Case Studies (SE, graduate level), University of Salzburg |
| WS 22/23 | Seminar Multimedia Technologies (SE, graduate level), University of Salzburg |
| WS 22/23 | Computer Vision (VO+PS, graduate level), University of Salzburg |
| SS 22 | Statistical Learning Theory (VO+PS, graduate level), University of Salzburg |
| SS 22 | Databases 1 (VO+PS, undergraduate level), University of Salzburg |
| SS 22 | Medical Imaging (VO+PS, graduate level), University of Salzburg |
| SS 22 | Seminar Multimedia Technologies (SE, graduate level), University of Salzburg |
| WS 21/22 | Interpreting and Presenting Statistical Analyses (SE, graduate level), University of Salzburg |
| WS 21/22 | Case Studies (SE, graduate level), University of Salzburg |
| WS 21/22 | Introduction to Data Science (SE, graduate level), University of Salzburg |
| WS 21/22 | Seminar Multimedia Technologies (SE, graduate level), University of Salzburg |
| WS 20/21 | Computer Vision (VO+PS, graduate level), University of Salzburg |
| SS 21 | Statistical Learning Theory (VO+PS, graduate level), University of Salzburg |
| SS 21 | Databases 1 (VO+PS, undergraduate level), University of Salzburg |
| SS 21 | Imaging Beyond Consumer Cameras (VO+PS, graduate level), University of Salzburg |
| SS 21 | Seminar Multimedia Technologies (SE, graduate level), University of Salzburg |
| WS 20/21 | Interpreting and Presenting Statistical Analyses (SE, graduate level), University of Salzburg |
| WS 20/21 | Case Studies (SE, graduate level), University of Salzburg |
| WS 20/21 | Introduction to Data Science (SE, graduate level), University of Salzburg |
| WS 20/21 | Seminar Multimedia Technologies (SE, graduate level), University of Salzburg |
| WS 20/21 | Computer Vision (VO+PS, graduate level), University of Salzburg |
| SS 20 | Statistical Learning Theory (VO+PS, graduate level), University of Salzburg |
| SS 20 | Databases 1 (VO+PS, undergraduate level), University of Salzburg |
| SS 20 | Imaging Beyond Consumer Cameras (VO+PS, graduate level), University of Salzburg |
| SS 20 | Seminar Multimedia Technologies (SE, graduate level), University of Salzburg |
| WS 19/20 | Interpreting and Presenting Statistical Analyses (SE, graduate level), University of Salzburg |
| WS 19/20 | Case Studies (SE, graduate level), University of Salzburg |

WS 19/20 Introduction to Data Science (SE, graduate level), University of Salzburg
WS 19/20 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg
WS 19/20 Computer Vision (VO+PS, graduate level), University of Salzburg
SS 19 Imaging Beyond Consumer Cameras (VO+PS, graduate level), University of Salzburg
SS 19 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg
SS 19 Statistical Learning Theory (VO+PS, graduate level), University of Salzburg
SS 19 Databases 1 (PS, undergraduate level), University of Salzburg
WS 18/19 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg
WS 18/19 Computer Vision (VO+PS, graduate level), University of Salzburg
WS 18/19 Introduction to Data Science (SE, graduate level), University of Salzburg
WS 18/19 Case Studies (SE, graduate level), University of Salzburg
WS 18/19 BSc Seminar (SE, undergraduate level), University of Salzburg
SS 18 Imaging Beyond Consumer Cameras (VO+PS, graduate level), University of Salzburg
SS 18 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg
SS 18 Statistical Learning Theory (VO+PS, graduate level), University of Salzburg
SS 18 Databases 1 (PS, undergraduate level), University of Salzburg
SS 18 BSc Seminar (SE, undergraduate level), University of Salzburg
WS 17/18 Wissenschaftliche Arbeitstechniken (VP, undergraduate level)
WS 17/18 Image Processing and Computer Vision (VO+PS, undergraduate level), University of Salzburg
WS 17/18 Case Studies (SE, graduate level), University of Salzburg
WS 17/18 Computer Vision (VO+PS, graduate level), University of Salzburg
WS 17/18 Introduction to Data Science (SE, graduate level), University of Salzburg
SS 17 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg
SS 17 Statistical Learning Theory (VO+PS, graduate level), University of Salzburg
SS 17 Computer Science for Everyone (VO, undergraduate level), University of Salzburg
SS 17 Databases 1 (VO+PS, undergraduate level), University of Salzburg
SS 17 BSc Seminar (SE, undergraduate level), University of Salzburg
WS 16/17 Introduction to Operating Systems (VO, undergraduate level), University of Salzburg
WS 16/17 Computer Vision (VO+PS, graduate level), University of Salzburg
WS 16/17 Introduction to Data Science (SE, graduate level), University of Salzburg
WS 16/17 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg
SS 16 Imaging Beyond Consumer Cameras (VO+PS, graduate level), University of Salzburg
SS 16 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg
SS 16 Databases 1 (PS, undergraduate level), University of Salzburg
WS 15/16 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg
WS 15/16 Advanced Image Processing & Computer Vision (VO+PS, graduate level), University of Salzburg
SS 15 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg
SS 15 Databases 1 (PS, undergraduate level), University of Salzburg
SS 15 Statistical Learning Theory (VO+PS, graduate level), University of Salzburg
WS 14/15 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg
WS 14/15 Advanced Image Processing & Computer Vision (VO+PS, graduate level), University
SS 14 Databases 1 (PS, undergraduate level), University of Salzburg
SS 14 Imaging Beyond Consumer Cameras (VO+PS, graduate level), University of Salzburg
SS 14 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg
WS 13/14 Advanced Image Processing & Computer Vision(VO+PS, graduate level), University of Salzburg
WS 10/11 Introduction to Object Oriented Programming (VO, undergraduate level), FH Salzburg
WS 05/06 Network Management (UE, undergraduate level), FH Salzburg

## Service to the University of Salzburg

| | |
|---|---|
| 2022 - now | Head of the *Curricularkommission BA Artificial Intelligence* |
| 2022 - now | Head of the division *Computer Vision and Machine Learning (CVML)* |
| 2022 - now | Deputy head of the department of *Artificial Intelligence and Human Interfaces (AIHI)* |
| 2016 - 2024 | Member of the *Curricularkommission MA Data Science* |
| 2019 - 2021 | Member of the *Curricularkommission BA/MA Computer Science* |
| 2019 | Appointment committee member for the *Assistant Professor position in Database Systems* |
| 2017 - 2018 | Appointment committee member for the *§ 99 UG professorship for Data Science* ("Stiftungsprofessur") |

## Professional Service

Area Chair (AC) for ICML '22, ICML '23, ICML '24, ICML '25, ICML '26, NeurIPS '21, NeurIPS '22, NeurIPS '23, NeurIPS '24, NeurIPS '25, NeurIPS '26, AAAI '25, ICLR '25, ICLR '26
General Chair of the 39th OAGM/AAPR Workshop 2015, Salzburg, Austria
Organizer of the 3rd Biannual Austrian TDA Meeting 2022, Salzburg, Austria
PC Chair of ACM IH & MMSEC 2014, Salzburg, Austria

### Journal reviewing

Reviewer for *Journal of Machine Learning Research (JMLR)*
Reviewer for *IEEE Transactions on Medical Imaging (TMI)*
Reviewer for *IEEE Transactions on Image Processing (TIP)*
Reviewer for *IEEE Transactions on Signal Processing*
Reviewer for *IEEE Signal Processing Letters*
Reviewer for *Elsevier Medical Image Analysis (MedIA)*
Reviewer for *Foundations of Computational Mathematics (FOCM)*
Reviewer for *Journal of Applied and Computational Topology (APCT)*

### Conference reviewing

Reviewer for *International Conference on Learning Representations (ICLR)*
Reviewer for *International Conference on Machine Learning (ICML)*
Reviewer for *Artificial Intelligence and Statistics (AISTATS)*
Reviewer for *Neural Information Processing Systems (NeurIPS)*
Reviewer for *IEEE International Conference on Image Processing (ICIP)*
Reviewer for *Medical Image Computing and Computer Assisted Intervention (MICCAI)*
Reviewer for *International Conference on Computer Vision (ICCV)*
Reviewer for *Computer Vision and Pattern Recognition (CVPR)*
Reviewer for *European Conference on Computer Vision (ECCV)*
Reviewer for *British Machine Vision Conference (BMVC)*
Reviewer for *International Conference on Pattern Recognition (ICPR)*

### Funding agencies

Reviewer for the *European Research Council (ERC)*

## References

**Prof. Marc Niethammer**
University of North Carolina (Chapel Hill)
250 Brooks Building
Chapel Hill, NC 27599-3175, USA
☎ (919) 590-6149
✉ mn@cs.unc.edu

**Stephen Aylward**
Kitware Inc.
101 East Weaver Street, Suite G4
Carrboro, NC 27510, USA
✉ stephen.aylward@kitware.com

**Prof. Nuno Vasconcelos**
University of California (San Diego)
EBU 1, Room 5603
9500 Gilman Drive, La Jolla, CA 92093-0407, USA
☎ (858) 534-5550
✉ nuno@ucsd.edu

**Prof. Baba Vemuri**
University of Florida (Gainesville)
Room No. E324, CSE Building
Gainesville, Florida 32611-6120, USA
☎ (352) 294-6675
✉ vemuri@ufl.edu

*More references available upon request.*

Last updated: April 6, 2026